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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/591,025	02/02/2022	Rémy Orban	2019P00556 US (G146US2)	2893

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EXAMINER	
HAWKINS, DOMINIC E	

ART UNIT	PAPER NUMBER
2858	

NOTIFICATION DATE	DELIVERY MODE
07/28/2023	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

Patent-Docketing@schneider-electric.com
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Office Action Summary	Application No. 17/591,025	Applicant(s) Orban et al.	
	Examiner DOMINIC E HAWKINS	Art Unit 2858	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on April 18, 2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-18 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-18 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 04/18/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Response to Amendments

2. Claims 1-18 of U.S. Application 17/591,025 filed on 04/18/2023 are presented for examination.

Response to Arguments

Entry of Amendments

3. Amendments to claim 14 has been entered.

Rejections under USC 102 and 103

Applicant's arguments filed on 04/16/2014 have been fully considered but are moot in view of a new rejection.

Claim Rejections - 35 USC § 112

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:

The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

4. Claims 1, 10, and 18 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as failing to set forth the subject matter which the inventor or a joint inventor, or for pre-AIA the applicant(s) regard as their invention.

5. Regarding claims 1 ,10, and 14, the phrases "while the electromagnetic actuator is kept in a stable position, in particular a closed position" are indefinite because it is unclear whether the limitations following the phrase are part of the claimed invention. See MPEP § 2173.05(d). It is unclear if the actuator is really required to be in a closed position, or if the actuator could be in a "stable position". The Examiner will interpret "while the electromagnetic actuator is kept in a stable position, in particular a closed position" as the actuator is in a closed position.

6. Regarding claim 10, the phrases "...a time interval...," "...means of sensor...," and "...a stable position..." are indefinite because it is unclear whether the limitations following the phrase are part of the claimed invention. See MPEP § 2173.05(d). It is unclear if these elements are the same or different than the elements in claim 1 because they are all recited in claim 1. The Examiner will interpret "...a time interval...," "...means of sensor...," and "...a stable position..." as the same elements that are in claim 1.

7. Appropriate corrections are required. Claims 2-9, 11-13, and 15-18 are also rejected as they depend on claims 1, 10, and 14 respectively.

Claim Rejections - 35 USC §101

22. 35 U.S.C. 101 reads as follows:

Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

23. Claims 1-18 are rejected under 35 U.S.C. 101 because the claimed invention is directed to a judicial exception (i.e., a law of nature, a natural phenomenon, or an abstract idea) without significantly more.

Claims 1 and 14 are directed to an abstract idea of using a mathematical function/algorithm for estimating the state of an electrical switching apparatus, recited in the limitations of "computing, by means of an electronic control device of the switching apparatus, one or more values of a magnetic flux of the coil from the measured coil current values, and using previously known coil voltage and coil resistance values.."

The claim(s) does/do not include additional elements that are sufficient to amount to significantly more than the judicial exception because the additional elements (ie, contacts, coil, sensors, electronic control device) (2)) are recited at a high level of generality in order to facilitate the application of the abstract idea and are recited as performing generic computer functions routinely used in computer applications. The collective functioning of the elements listed above merely provide conventional computer implementation. **The independent claims does not have a limitation to where the data is further used to determine properties such as anomalies, damage location of the switching apparatus which would give the claims significantly more.**

Dependent claims 2-13 and 15-18 when analyzed as a whole are held to be patent ineligible under 35 U.S.C. 101 because there is no additional element(s) in the dependent claim(s) that adds a meaningful limitation to the abstract idea to make the claim significantly more than the judicial exception (abstract idea).

Please refer to court cases listed below for reference to the abstract ideas having a similar concept in the field of electronics:

Electric Power Group, West view: Collecting information, analyzing it, and displaying certain results of the collection and analysis.

Content Extraction: Data recognition and storage.

Digitech: Organizing and manipulating information through mathematical correlations.

Grams: An algorithm for calculating parameters indicating an abnormal condition.

The concept of these court cases are obtaining data, manipulating information through mathematical correlations, analyzing and displaying data which all refers to an abstract idea which would be performed by a generic computer.

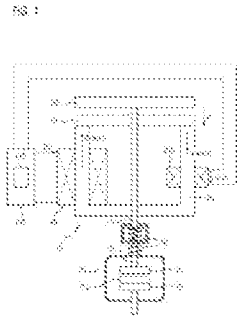
Claim Rejections - 35 USC § 103

1. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

2. Claim 1 is rejected under 35 U.S.C. 103 as being unpatentable over Hahn et al (USPGPub 20160293364) in view of Corbetta et al (USPGPub US Pat No. 6859350).



Prior Art: Hahn

3. Regarding claim 1, Hahn discloses a method for estimating the state of an electrical switching apparatus (20), the switching apparatus comprising separable electrical contacts set up

to be coupled to an electrical conductor (not fully shown but I(t) comes from 100 therefore there has to be a conductor and contacts to the coil. Also since this is a method claim and these components does not necessarily involve in the steps/method of performing testing), and an electromagnetic actuator (10), controlled by a control circuit (110) and comprising a movable core (60 including 50) , coupled to the separable contacts, and a coil (80 and 200) , which is passed through by a coil current (It), wherein the method comprises: measuring, by means of sensors (using Hall 400) associated with the coil, values of the coil current during a time interval (time profile), while the electromagnetic actuator is kept in a stable position (such as position 61), in particular a closed position (par 33 discloses being in closed position); computing, by means of an electronic control device (100) of the switching apparatus, one or more values of a magnetic flux of the coil from the measured coil current values (claim 1 and abstract discloses at least voltage). Hahn does not fully disclose using previously known coil voltage and coil resistance values.

However, Corbetta discloses using previously known coil voltage and coil resistance values (col 5 lines 15-40 discloses previous calculations Vc and Rc are known values. Therefore, uses these values to obtain data). It would have been obvious to a person having ordinary skill in the art at the time the invention was made to have provided combined Hahn in view of Donderici in order to determine characteristics of an object by calculating actual vs stored information.

Allowable Subject Matter

4. Claims 2-13 and 15-18 would be allowable if rewritten to overcome the rejection(s) under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), 2nd paragraph, and 35 U.S.C 101, set forth

in this Office action and to include all of the limitations of the base claim and any intervening claims.

5. Regarding claim 2, the prior art of record taken alone or in combination fail to teach or suggest a method for estimating the state of an electrical switching apparatus, the switching apparatus comprising separable electrical contacts set up to be coupled to an electrical conductor, and an electromagnetic actuator, controlled by a control circuit and comprising a movable core, coupled to the separable contacts, and a coil, which is passed through by a coil current wherein the method comprises wherein the one or more values of a magnetic flux ~~(0)~~ are computed by integrating the following formula:

$$\phi = \int \frac{U_{BOB} - R_{BOB} \cdot I_{BOB}}{N_{BOB}} dt \qquad \text{[Math 5]},$$ where 0 is the magnetic flux, UBOB is the previously known coil voltage, RBOB is the coil resistance, IBOB is the coil current, and NBOB is a number of turns of the coil in combination with the other limitations of the claim.

6. Regarding claim 3, the prior art of record taken alone or in combination fail to teach or suggest a method for estimating the state of an electrical switching apparatus, the switching apparatus comprising separable electrical contacts set up to be coupled to an electrical conductor, and an electromagnetic actuator, controlled by a control circuit and comprising a movable core, coupled to the separable contacts, and a coil, which is passed through by a coil current wherein the method comprises wherein a first value of the magnetic flux is computed from measurements of the coil current taken while the switching apparatus is in a freewheeling mode, in which the

coil voltage is passively limited by the control circuit, while the coil current decreases freely, with the electrical energy being mainly dissipated by the coil resistance in combination with the other limitations of the claim.

7. Claims 4-7 are also objected to as they depend on objected claim 3.

8. Regarding claim 8, the prior art of record taken alone or in combination fail to teach or suggest a method for estimating the state of an electrical switching apparatus, the switching apparatus comprising separable electrical contacts set up to be coupled to an electrical conductor, and an electromagnetic actuator, controlled by a control circuit and comprising a movable core, coupled to the separable contacts, and a coil, which is passed through by a coil current wherein the method comprises wherein the one or more computed values of magnetic flux are normalized and combined with at least one other measured and/or computed electrical quantity of the switching apparatus, with each other electrical quantity measurement also being normalized, so as to arrive at an intermediate indicator of the state of the electromagnetic actuator in combination with the other limitations of the claim.

9. Claims 9-13, and 15-18 are also objected to as they depend on objected claim 3.

10. Regarding claim 14, the prior art of record taken alone or in combination fail to teach or suggest an electrical switching apparatus comprising: separable contacts, configured to be moved between an open position and a closed position by an electromagnetic actuator comprising a coil

and a movable core linked to the separable contacts; a control circuit configured to control electrical voltage at ~~the~~ terminals of the coil, the control circuit comprising a fallback device configured to be selectively activated to reduce ~~the~~ electrical current passing through the coil, with the voltage at the terminals of the coil and the fallback device being activated or deactivated according to states of the control circuit; sensors associated with the coil for measuring the current passing through the coil; an electronic control device, configured to receive opening and closing commands from the switching apparatus, to receive values of the measured current passing through the coil and to control the states of the control circuit in combination with the other limitations of the claim.

Prior Art

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.
Donderici et al (USPGPub 20160258276): discloses using a processor type device uses previously determined voltage and impedance.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to DOMINIC E HAWKINS whose telephone number is (571)272-2647. The examiner can normally be reached Monday-Friday 7:30am-5:00pm EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Judy Nguyen can be reached on (571) 272-2258. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/DOMINIC E HAWKINS/

Primary Examiner, Art Unit 2858